

Manufacturing Production-Decision Capability Development

SOLUTION WHITEPAPER

Starting from real data scenarios, build production decision capabilities that can be run, reviewed, integrated, and sustainably evolved.

DecisioWorks

PRODUCTION DECISION TOOLKIT

DATA / MODEL / SOLVER / DECISION



Version 1.2

September 2026

Solution documentation

DECISIOWORKS SOLUTION WHITEPAPER

SOLUTION OVERVIEW

DecisioWorks supports manufacturing companies to start from specific scenarios, connect data preparation, operational validation, system integration and continuous operations, and gradually build operational and reviewable production decision capabilities.

project	illustrate
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Scope of application

It is suitable for the construction of production decision capabilities of manufacturing enterprises, as well as the development of industry solutions for software manufacturers, consulting agencies and system integrators, covering scenario selection, data preparation, operational validation and ecosystem integration.

System and organization collaboration

DecisioWorks provides auxiliary decision capabilities: the system generates plans, explains differences and presents impacts, and business personnel complete review, selection and execution arrangements based on on-site conditions.

The article uses public examples for injection molding, stamping and other content to show how data, models, solutions, results and feedback form a complete link, making it easier for readers to understand the methods and carry out scenario exercises.

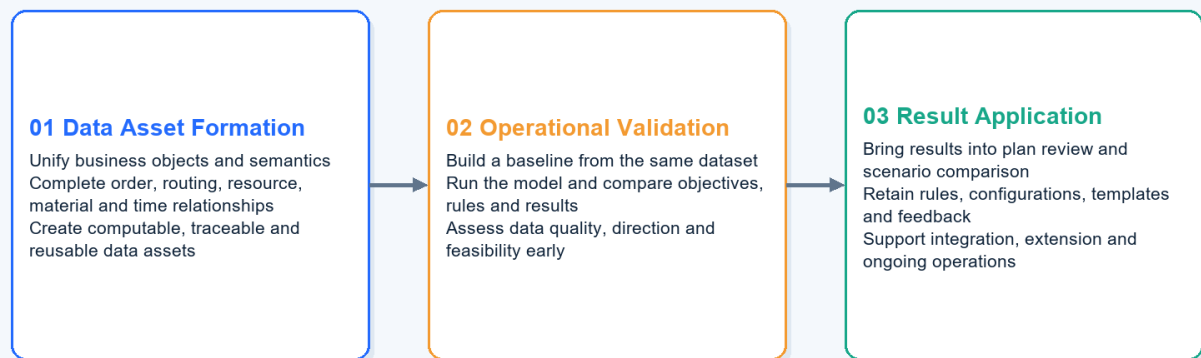
DECISIOWORKS SOLUTION WHITEPAPER

EXECUTIVE SUMMARY

The value of DecisioWorks solution can be summarized as: starting from real data scenarios, transforming dispersed production information into production decision capabilities that are runnable, reviewable, integrated and capable of continuous evolution.

From Data Assets to Operational Results

Three value levels on one continuous path



Form the asset → validate the asset → use the asset

DecisioWorks

Executive Summary Diagram: Data Asset Formation, Operational Validation and Results Application

key questions	Corresponding capabilities and application methods
There is a lot of data but it cannot be scheduled directly	Build computable data assets with standard business objects, data specifications and five levels of readiness
The plan has been generated, but there is insufficient basis for adoption.	Generate decision evidence through goal rules, controlled solving, object drill-down, and manual review
The project cycle is long and ROI is unclear	Establish a baseline with the minimum scenario, and determine subsequent investment based on the operation and review results
Already have ERP/MES/WMS but lack decision ability	Form an embeddable enhancement layer with open interfaces, DecisioCore, GOCK and adapters
Insufficient reuse across projects	Retain data templates, standard actions, scenario recipes and model adaptation into reusable assets

solution value

DecisioWorks does not require companies to build a huge system first, but helps companies and partners start from a real scenario and gradually establish a complete link that transforms production data into plans, judgment basis and feedback capabilities.

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System and organization collaboration

The system is responsible for organizing data, generating candidate solutions and presenting differences; planning, production and management personnel combine on-site conditions to confirm target selection, execution arrangements and feedback requirements. Computing capabilities and business judgment are interconnected, allowing planning results to enter the actual decision process of the enterprise.

System and Organization Decide Together

The system supplies computable evidence; people make accountable choices

System: Compute & Explain

- Validate data and relationships
- Express objectives and operating rules
- Run controlled optimization
- Compare scenarios and diagnose conflicts
- Trace results and feed back deviations

Joint
Decision

Organization: Review & Commit

- Set priorities and trade-offs
- Confirm site-level feasibility
- Commit to delivery, cost and risk
- Approve exceptions and coordinate resources
- Adopt, dispatch and review outcomes

A reviewable, accountable and executable production decision

DecisioWorks

The system and organization collaborate to complete production decisions

DECISIOWORKS SOLUTION WHITEPAPER

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SECTION 01

01 Industry background: production decision enters the capacity building stage

Intelligent manufacturing is moving from equipment connection, digital business operations and data aggregation to plan optimization, dynamic scheduling and collaborative decision. production decision is no longer just an ancillary function in a certain project, but has begun to become a basic capability that manufacturing companies need to build and operate in the long term.

The Ministry of Industry and Information Technology's "Reference Guidelines for Typical Scenarios of Intelligent Manufacturing (2025 Edition)" lists production plan optimization, intelligent scheduling, etc. as typical scenarios, emphasizing multi-objectives, multi-constraints, cross-link data collaboration and dynamic adjustment. This shows that the industry's focus has shifted from "whether there is a system" to "whether data can enter decision, whether plans can be dynamically formed, and whether results can be continuously used." [1]

Digital transformation policies for small and medium-sized enterprises continue to emphasize modularity, lightweight, low cost, easy deployment, open interfaces and replicable promotion. production decision capabilities can be built through phased verification, integration, and expansion to gradually form reusable capabilities in clear scenarios. [2][3]

Key issues in production decision and construction

The company already has ERP, MES, WMS and a large number of tables. Why does the production plan still rely heavily on manual labor?

How do data, goals, rules, models, solutions, and feedback form an operational decision chain?

How to use a scenario with clear boundaries to first verify data, plans and investment directions?

How can industrial software partners complement production decision capabilities without building the entire APS from scratch?

How to let the system handle the computable part and support the organization in completing plan selection and execution arrangements?

Construction focus

The key to building production decision capabilities is not to add a scheduling page.

Instead, data, business, models, solutions, results, and feedback are organized into links that can be verified, reviewed, and continuously used.

SECTION 02

02 Key requirements for building production decision capabilities

production decision solutions need to cover business problems, constraint modeling, planning and scheduling, scenario analysis, system integration and result adoption at the same time. The public materials of SAP PP/DS, Oracle Production Scheduling, Siemens Opcenter APS, and DELMIA Ortems demonstrate the application of these capabilities in different products. [5][6][7][8]

When companies evaluate plans, in addition to paying attention to scheduling results, they also need to confirm how resource and material conditions are expressed, how changes trigger recalculation, how planners adjust and review, and how the results are connected to existing business systems. The following table lists the representative capabilities and application concerns of relevant public solutions.

public plan	representative ability	Enterprise application focus
SAP PP/DS	Key products, resource and component availability, detailed scheduling, heuristics and optimization, interaction planning	Collaboration between the planning layer and the scheduling layer, as well as interactive adjustments by planners
Oracle Production Scheduling	Capacity and material constraints, model changeover, delayed orders, scenario analysis, ERP/MES data connection	Comparison of resources, materials and plans under changed conditions
Siemens Opcenter APS	Limited capacity, scheduling constraints, manufacturing operation connection, and plan visualization	The connection between planning and scheduling and manufacturing execution system
DELMIA Ortems	Demand-supply-production synchronization, dynamic rearrangement, KPI, planner interaction	Recalculation, indicator comparison and business review after on-site changes

DecisioWorks's ability organization method

DecisioWorks organizes standard data access, goals and rules, capability orchestration, model solving, result analysis and feedback into an integrable toolkit. Enterprises and partners can combine these capabilities on the basis of existing systems and gradually form production decision applications suitable for their own industry processes.

Key points for program evaluation

A complete capacity building plan should clarify data sources, rule expressions, result review, system integration, acceptance criteria and continuous maintenance methods.

Enable technical capabilities to enter daily production management.

SECTION 03

03 Why is production decision difficult?

Production decisions seem to be answering "what to produce, when to produce and where to produce", but are actually affected by demand, process, resources, materials, time, delivery, cost, stability and on-site changes. It is not a single algorithm problem, nor a single system problem.

Overlay of multiple types of complexity

complexity	Typical performance	easy consequences
Data complexity	The same order, product or resource has different definitions in multiple systems	Data can be imported, but relationships cannot be closed
structural complexity	Routing, alternative resources, batch size, shift, maintenance and kiting interact with each other	Model misses key conditions or scales out of control
Goal complexity	Delivery time, inventory, model change, load, and stability cannot be infinitely improved at the same time	The system gives "optimal" results, but the organization does not agree with them
changing complexity	Inserting orders, shortage of materials, failures, changes in delivery time and personnel experience continue to enter	After business changes, the original configuration may no longer be applicable.
Responsibility Complexity	Results impact customer commitments, costs, personnel and contractual responsibilities	The system cannot bear the final consequences for the organization

Therefore, the production-decision system must deal with both "calculus" and "adoptability". The former requires data, constraints, goals and solution structures to be rigorous; the latter requires results to be interpreted, compared, reviewed, and entered into organizational processes. If you only satisfy one end of the spectrum, it will be difficult to create long-term value.

The connection between algorithms and business

Algorithmic ability is important, but algorithms only work within given variables, constraints, and goals.

Business semantics, on-site rules, responsibility boundaries and feedback mechanisms must be carried out by a complete capability chain.

SECTION 04

04 Typical pain point matrix

Different participants see different symptoms: manufacturing companies see plans being revised repeatedly, software vendors see customers constantly raising new demands, consulting agencies see solutions being difficult to verify, and integrators see interfaces and customizations continue to expand. The root cause often points to the same decision chain not being closed.

Role	Common pain points	Abilities that need to be supplemented
manufacturing company	Data is scattered, plans rely on individuals, results are difficult to interpret, ROI is unclear	Data assets, scenario verification, result review, continuous feedback
ERP/MES/WMS Manufacturer	Transaction and execution data are complete, but plan optimization capabilities are weak	Standard semantic mapping, embeddable solution, analysis and feedback interface
Industry software vendors	There are industry interfaces and processes, but there is a lack of a stable model-based decision base.	Capability registration, scenario recipe, model adaptation, authorized operation
consulting agency	Can make suggestions for improvement, but it is difficult to form operable and comparable evidence	Data diagnostics, baseline recalculations, scenario comparisons and assessment materials
system integrator	Repeatedly organizing data, writing rules, and docking solvers for each project	Reusable interfaces, standard actions, deployment and support boundaries
Manufacturing IT team	Want to build it yourself, but the model, engineering and long-term maintenance requirements are high	Open reference implementation, controlled kernel, layered extensions and documentation

The connection between pain points

Data semantics, rule expression, result interpretation, and execution feedback interact with each other. When data or rules are incomplete, manual adjustments and supplementary scripts may continue to increase. Unified organization of these links can help locate the source of the problem and reduce repeated corrections.

Scope of application description

The above scenarios are common application examples and are not limited to the full scope of application of DecisioWorks. Other scenarios can be further evaluated based on problem definition, data conditions, and results review requirements.

SECTION 05

05 production decision construction path and applicable conditions

Different construction paths are suitable for different business foundations and organizational conditions. If the data, models, and results are not verified until later in the implementation, the initial investment may face major adjustments. Confirming these conditions in stages can help identify risks early.

common paths	Applicable advantages	Conditions and risks that require attention
Comprehensive APS implementation	Can cover more processes and interfaces	Need to coordinate integration scope, implementation cycle and maintenance after business changes
Rules and script implementation	Close to current experience, quick results	Requires management of rule sources, conflict checking and knowledge transfer
Single point algorithm interface	Quickly display optimization results	Need to supplement data semantics, business context and result application links
Reports and Kanban	Data is visible and communication is intuitive	Plan generation and business selection still require corresponding capability support
Completely self-developed	Independent, controllable and deeply customizable	Need to bear model, engineering, product and operation and maintenance costs at the same time

A large amount of customized logic will increase the maintenance burden, and optimization results lacking explanation materials will be difficult to gain business acceptance. Hierarchical management of data, goals, rules, orchestration and models allows changes to be processed at the corresponding levels while retaining traceable operational basis.

Phased application method

Start with a data scenario with clear boundaries rather than covering the entire factory first.

First form data and operational evidence, and then decide whether to expand investment.

The toolkit provides common capabilities, and partners undertake industry processes, system integration and delivery services.

Allow existing systems to be enhanced without requiring replacement of the entire business foundation.

SECTION 06

06 Solution positioning and boundaries

DecisioWorks is an open capability toolkit for manufacturing production planning, job scheduling and collaborative decision. It helps manufacturing companies and ecosystem partners organize production data into operational decision capabilities, and supports the gradual move from partial verification to system integration and continuous operations.

Provided by DecisioWorks	Undertaken by enterprises or partners
Standardized data interface, semantic definition and data readiness check	Source system governance, data responsibility, on-site definition confirmation
Goals, rules, DecisionContext and capability orchestration	Management trade-offs, rule sources, process design, and organizational approvals
GOCK controlled solving and model adaptation	Industry model selection, on-site parameter confirmation and business acceptance
Result analysis, evidence, deviation and feedback interface	Plan acceptance, plan distribution, implementation arrangements and effect review
Open reference implementation, documentation, samples and integration interfaces	Industry interface, customer relations, deployment, operation and maintenance and delivery services

DecisioWorks does not require companies to rebuild existing systems from scratch, nor does it require replacement of ERP, MES, WMS or partner applications. It can be used as a stand-alone validation tool, a decision-capability enhancement layer for existing systems, a consulting assessment tool, or an embedded capability node in a partnership program.

ecosystem principles

Ceju focuses on toolkit research and development, capability licensing and product support;
Ecosystem partners take advantage of industry knowledge, system integration and on-site services to organize the implementation and operation of customer applications.
Both parties support customers' long-term applications with a clear division of labor.

SECTION 07

07 Construction goals and design principles

The solution takes on site complexity through a layered architecture. When data, business rules, models and processes change, they can be adjusted at the corresponding level, and configurations, operating results and review records are retained.

design principles	meaning	Implementation method
Data is business	Simultaneously express business objects, relationships, and structural constraints as data enters	Standard interface, mapping, semantic verification
Planning and Decision Layering	Plans are formed by models and decisions are made by the organization based on evidence	GOCK, DecisioCore, manual review
Explicit goals and rules	Management preferences and field experience are not hidden in algorithmic code	objective_system、marginal_control
Capabilities can be programmed	Steady action can be recombined by scene	Capability registration, recipe, Pipeline/DAG/FSM
result forms evidence	Overall indicators, object details and audit materials are retained at the same time	Analysis, diagnostic ID, version history
Openness and evolution	Data, business, model, solution and interface can be expanded separately	DecisioCore, adapter, API, document

Three acceptance goals

Operable: With the same data and configuration, the link can perform stably and return to a clear state.

reviewable: The results can be drilled down to orders, resources, processes or time units, and the basis for formation is explained.

Scalable: Data, rules, recipes and interfaces that pass verification can enter the next scenario instead of starting from scratch each time.

solution idea

What is visible is the clear process, but what is invisible is the rigorous capabilities composed of data, models, rules, authorization and evidence.

SECTION 08

08 Overall solution architecture

DecisioWorks takes "Scenario × Business × Model × Solver" (Scene × Business × Model × Solver) as the top-level logic and is implemented in collaboration with DataSets, DecisioCore, GOCK and Web Cockpit. Each component is connected through data, call and feedback relationships to jointly support complete operation from scenario configuration to solution review.

Manufacturing Production-Decision Architecture

Business applications, decision orchestration, optimization and data assets work as one system

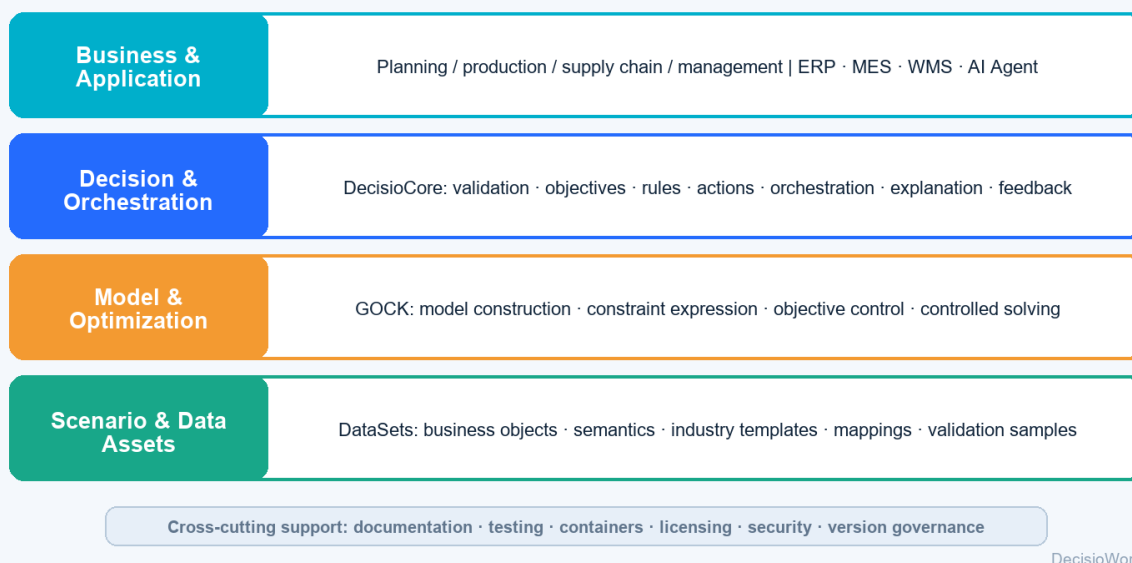


Figure 1 Overall structure of manufacturing production decision capacity building

Four types of core components

components	Main responsibilities	Application value
DataSets	Industry scenarios, standard data interface examples, data templates and samples	Allow business to be accurately expressed and reused
DecisioCore	Connect data, goals, rules, actions, results and feedback	Put plans into a reviewable decision process
GOCK	Model construction, mathematical expression of objectives and constraints, optimization calculations and controlled solving	Allow complex plans to form under clear conditions
Web Cockpit	Demonstrate inputs, configurations, processes, outputs and evidence	Make capabilities visible, actionable and verifiable

Documentation, tools, containers, testing, versions, security and authorization horizontally support the above levels. Each layer maintains collaboration through stable interfaces, while allowing data changes, business changes, and model upgrades to occur at the corresponding locations.

A typical run is: DataSets → DecisioCore (validation, target rules, orchestration) → GOCK (model and optimization) → DecisioCore (analysis, interpretation, feedback) → Web Cockpit / API / Next round of planning.

After the plan is reviewed by business personnel, it can enter the decision and execution process of the enterprise or partner organization.

SECTION 09

09 End-to-end production-decision capability chain

The capability chain connects data preparation, solution generation, organizational review and execution feedback. Each link has clear input and output, status and responsibility, which helps reduce manual supplementary processing and repeated customization.



Figure 2 End-to-end capability chain of production decision

link	key questions	main output
data entry	Are business objects and relationships accurate?	Standard input, mapping, ready state
decision expression	Are goals, rules and context clear?	Target material, rule list, DecisionContext
Capability orchestration	What actions does a run consist of?	Scenario recipes, execution status and audit records
controlled solving	Are the model and engine operating under permitted conditions?	Candidate solutions, running status, diagnostic information
Result review	Is the plan explainable, comparable and executable?	Overall indicators, object details, manual conclusions
feedback evolution	How deviations advance to the next round	Signals, rule adjustments, templates and versions

link value

Standard data, business configuration, model calling, result analysis and feedback interfaces are connected to each other, making the input, process and results of each run traceable.

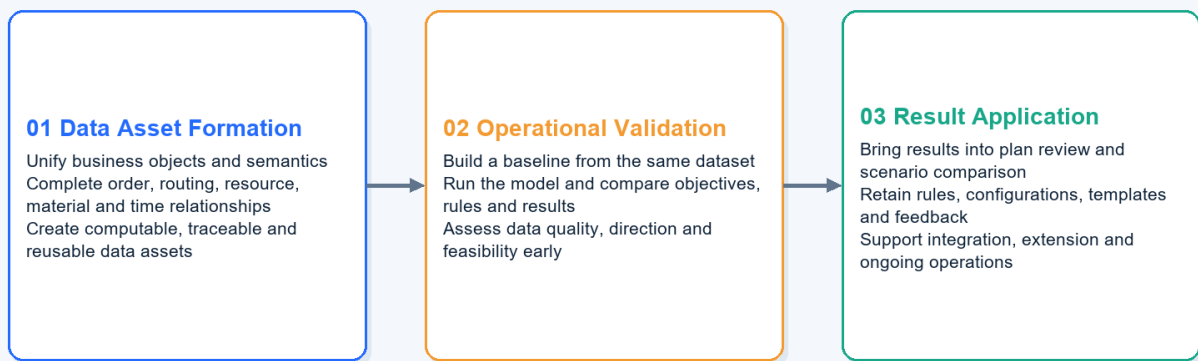
SECTION 10

10 Standardized data interfaces and decision data assets

Manufacturing companies usually do not have data, but the data has not yet formed stable business meaning, complete object relationships and clear quality status. Transforming scattered records into computable, traceable, and reusable data assets has itself generated value.

From Data Assets to Operational Results

Three value levels on one continuous path



Form the asset → validate the asset → use the asset

DecisioWorks

Figure 3 Three levels of value: data asset formation, operational validation and result application

production decision data assets include standard data templates, field mappings, object relationships, unit conventions, quality reports and running records. These reusable engineering and business data provide the basis for subsequent verification, integration and continuous decision.

The application value of data standardization

Clarify which systems are responsible for orders, products, processes, resources, materials and time.

Found the same name but different synonyms, synonyms but different names, unit conflicts and missing relationships.

Form handoff field mapping, data responsibilities and update methods.

Continue to test whether the data can support the plan and schedule through real operations.

Provides a common basis for subsequent system integration, model validation, training, and AI Agent invocation.

continuous value

Forming assets, verifying assets, and using assets are not three separate projects and can occur continuously in the same production decision chain.

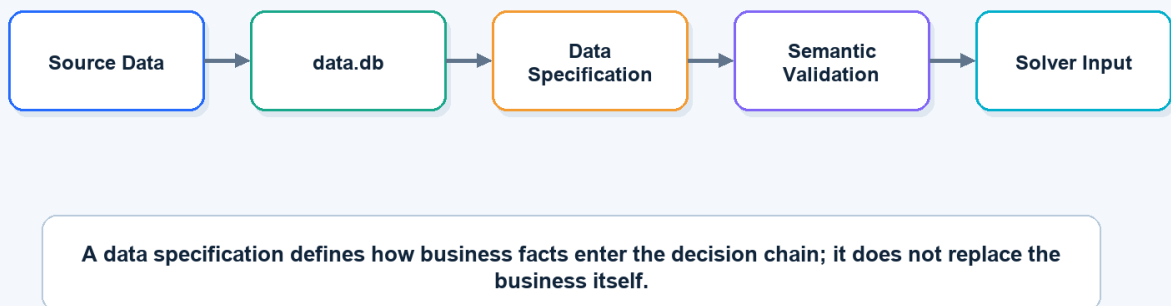
SECTION 11

11 ERP, MES, WMS mapping and data specification

Standardized access is not to mechanically copy the source system fields to a new table, but to first identify the business objects and relationships, and then decide where each field comes from, how to convert it, and who maintains it.

From data.db to a Computable Business Model

Data structure is only the start; business semantics and relationships must also close



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Figure 4 From Raw Data to a Data Specification and Solver Input

Four relationship chains that must be closed

relationship chain	Questions that must be answered	Typical errors
time chain	Whether delivery time, capacity, packaging and results use consistent time coordinates	Minutes, hours, shifts and time units are mixed
Product chain	Whether the order, route, inventory, and attributes point to the same article	Version or specification lost after encoding conversion
Process capacity chain	Can the product find executable resources through routes and processes?	Operation exists but no work center available
Material chain	Can process-level requirements correspond to materials, inventory or material readiness of information?	The BOM only reaches the product level and cannot determine that the processes are complete.

The data specification should retain the table structure, field semantics, units, primary keys, associations, required conditions, enumeration definitions, default rules and versions. It is not a document list, but an acceptance boundary for business semantics to enter the decision chain.

SECTION 12

12 Five-Level Data Readiness Model

The data can be read, which only proves that the files and fields are accessible. The five-level readiness status of structure, relationship, model, solution and decision is used for step-by-step acceptance: the first four levels check the conditions for entering solution, and DecisionReady combines the running results and business personnel for review and confirmation. The data can be solved and the results can be judged separately.

state	questions to answer	Typical evidence
ContractReady	Can tables, fields, types and identifiers be identified?	Structure report, required field check
RelationReady	Can the relationship between time, products, processes, resources and materials be closed?	Foreign Key and Relationship Chain Report
ModelReady	Can business facts be converted into model objects and structural constraints?	Capability assembly and constraint list
SolverReady	Can goals, rules, and configurations form a solvable problem?	Formula inspection, target material audit
DecisionReady	Can the results be interpreted, compared and confirmed by business personnel?	Object drill-down and plan review record

Readiness status and follow-up work

The readiness state is used to determine subsequent data preparation and verification efforts. When ContractReady is reached but RelationReady is not reached, priority should be given to repairing the object relationship; when ModelReady is reached but SolverReady is not reached, goals, rules or parameters should be added; when the solution can be solved but DecisionReady is not reached, result interpretation, business review and process access should be improved.

control principle

Missing data should be identified and supplemented. When it is necessary to use default values or business assumptions, it should be confirmed by the responsible person, and the source, applicable conditions and impact on the results should be recorded.

SECTION 13

13 DecisionContext, goals and rules

A production decision not only requires data, but also needs to explain what problems are to be solved in this round, which goals are used, which rules are effective, which tasks cannot be adjusted, and who will review the results. DecisionContext is used to run these conditions throughout the entire capability chain.



Figure 5 DecisionContext runs through input, constraints, candidate solutions and audit history

Content type	Answered questions	Carrying mode	Example
structural constraints	Can it be done objectively?	Standard data interface	Process sequence, capacity, shift, BOM, batch size
management objectives	Which plan is more preferred?	target system	Delivery, Load, Wait, Stock, Switch
site rules	How must the current scenario be handled?	marginal_control	Bottleneck protection, priority window, prohibition conditions, resource quota
DecisionContext	Why does this round run and who confirms it?	DecisionContext	Scenario, version, responsible person, scope of application, audit history

Mixing the three types of business content and context into the algorithm code will make the source of the rules unclear, changes difficult to maintain, and results unexplainable. When expressed separately, each adjustment can explain the target, source, validity period and impact.

SECTION 14

14 Standard actions, capability registration and orchestration recipes

The production decision process must be stable and allow for scenario differences. DecisioWorks splits data reading, constraint analysis, goal construction, solution generation, selection, evaluation and output into standard actions, and then combines them into scenario processes through recipes.

Capability Recipe: Turning Industry Know-how into a Runnable Flow

A recipe combines registered capabilities, action order, parameters and acceptance criteria



DecisioWorks

Figure 6 From Scenario Intent to Runnable Capability Orchestration

Three typical execution methods

Way	Suitable for the scene	Management focus
Pipeline	Stable sequence of steps, relying on a simple planning process	Input, output and stop on failure
DAG	There are parallel checks, multi-source dependencies and convergence nodes	Dependencies, local failures and retries
FSM	Driven by status changes, requiring manual confirmation or external events	State migration, conditions and auditing

The capability registry describes the input, output, status, parameters and applicable scope of the action. Web, script and AI Agent call the same set of registered capabilities to avoid each entry point implementing a set of hidden logic.

The toolkit provides orchestration mechanisms, standard actions and sample recipes; enterprises or partners are responsible for customer-specific processes, industry templates and external system connections.

Reuse unit

Reusable scenario recipes include scenario descriptions, data premises, action combinations, target rules, operational acceptance and failure conditions, and support inspection and adaptation during cross-scenario migration.

SECTION 15

15 GOCK: controlled solving and model adaptation

GOCK is a general-purpose optimization calculation core that handles model construction, optimization and controlled solving. The target system and rule modules of DecisioCore organize the business intent, which engine_adapters maps to the model call of GOCK. Business configuration, model building, and algorithm execution each accept changes and jointly support the formation of plans under clear conditions.

Model-Data Resonance

In Model-Data Resonance, "model" refers to the decision model and "data" refers to business data. Data carries orders, processes, resources, materials and time semantics into the model, and the model takes over on-site relationships, target preferences and constraints in a computable structure. The running results are then used for business review, data inspection and configuration adjustment, so that the model and data are continuously calibrated in the same capability chain. This synergistic mechanism is called "Model-Data Resonance."

control phase	Main inspection	The status that needs to be returned
Before running	Data set, engine type, target direction, rule legality, resource limit	Allow the run or specify the reason for denial
Running	Request identification, action status, time taken, cancellation or timeout	In progress, canceled, timed out or component exception
After running	Candidate solutions, selected solutions, result shapes, target directions and analysis materials	Successful, no feasible solution, available but no further improvements have been completed, etc.

engine_adapters is located between DecisioCore and GOCK. It converts upper-level business objects, goals and rules into model calls, and then restores running status and result semantics to upper-level analyzable structures. When the underlying model is upgraded, as long as the external contract remains stable, the business layer and application interface do not need to be rewritten simultaneously.

controlled solving

Timeout, no feasible solution, and program exception are different states; weights, resources, and authorizations also have clear scopes. Rigorous status expression is an integral part of reliable use in production decisions.

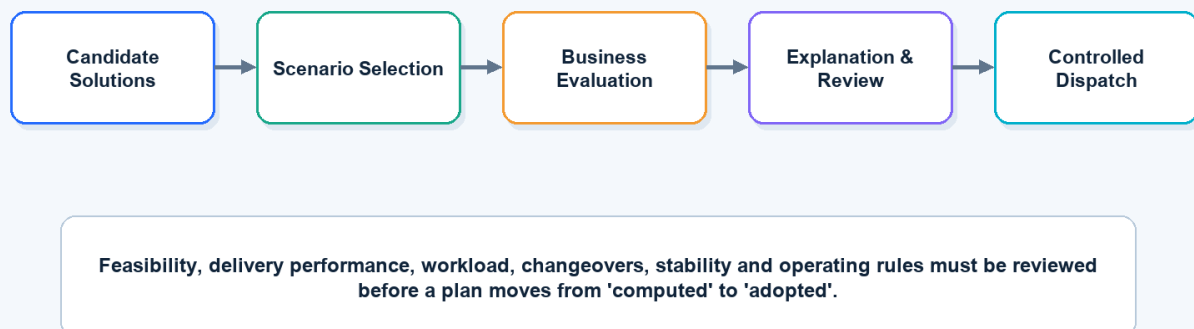
SECTION 16

16 Result Evidence, Interpretation and Manual Review

A schedule only shows that the model gives an arrangement and cannot alone prove that the plan is suitable for execution. Business people also need to know how much is satisfied, which objects are affected, what cost the improvement is, and whether to accept the next step.

From Candidate Solutions to a Usable Decision

A solver output becomes useful only after selection, business evaluation, review and controlled dispatch



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Figure 7 From candidate solutions to available decisions

result hierarchy	Answered questions	typical content
overall layer	How is the overall performance of the program?	Satisfaction, shortage, output, load, target profile
object layer	where the problem occurs	Work center, order, product, operation, time unit, switch
Audit layer	Whether the result is produced as configured	Data version, target direction, material mapping, parameters, status

The system should present both actual business volume and target scores. Weighted target values are used to compare trade-offs, minutes, quantities, loads and shortages must be read through the corresponding indicators. Mixing the two will lead to misjudgments such as "the score becomes better but the business meaning is unclear".

Business review

High-impact actions such as increasing or decreasing capacity, overtime, outsourcing, customer commitments, material replacement, and bottleneck intervention should be reviewed and approved by the designated responsible person of the enterprise.

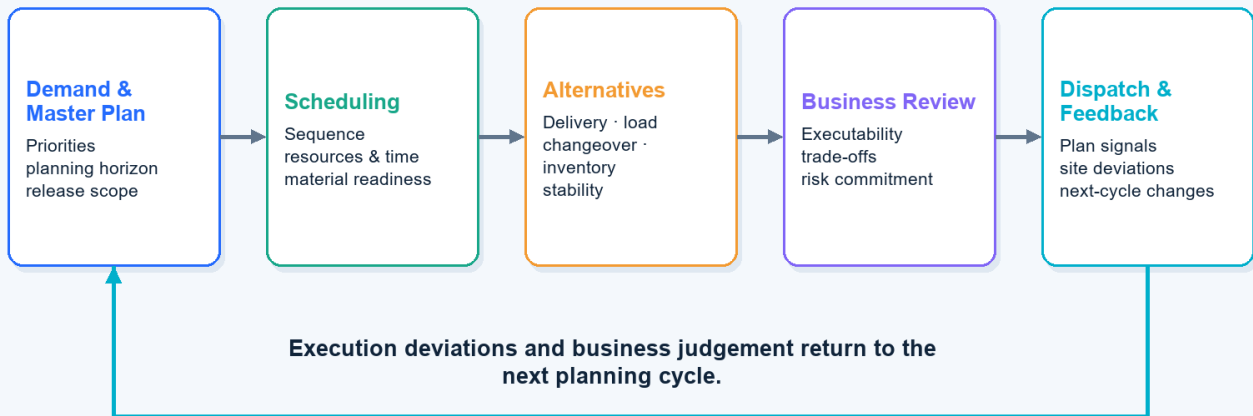
SECTION 17

17 From planning to making decisions

High-quality planning is an important basis for production decisions, but it is not the end point. Real decision also includes option comparisons, risk trade-offs, organizational reviews, controlled dispatch, and execution feedback.

From a Good Plan to a Good Decision

A plan becomes a production decision through comparison, review, dispatch and feedback



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Figure 8 From plan formation to organizational decision and feedback

GOCK is responsible for forming high-quality plans under clear data, constraints and goals; DecisioCore is responsible for organizing goals, rules, actions, explanations, evidence and feedback; Web Cockpit makes inputs, configurations, processes and results visible; DataSets provides industry scenarios and data templates. The four work together to advance the "plan results" into "decision materials that can be adopted by the organization."

From plan generation to solution adoption

level	core issues	Key points for acceptance
Make plans	How to arrange it more reasonably under given conditions	Feasibility, target performance, resource and material relationships
Make good decisions	Whether the organization understands, adopts and assumes the implementation commitment	Explain, compare, review, distribute, give feedback

core positioning

The system provides the basis for calculation and analysis, organizes decisions based on on-site conditions, and promotes implementation. The value of DecisioWorks is to reduce the information cost, calculation cost and review cost of organizational judgment.

SECTION 18

18 Plan linkage and feedback evolution

The production site continues to change, and stable capabilities cannot rely on one-time configuration. Data, constraints, impacts, and feedback need to form a closed loop, allowing valid experiences to advance to the next round and failure conditions to trigger revalidation.

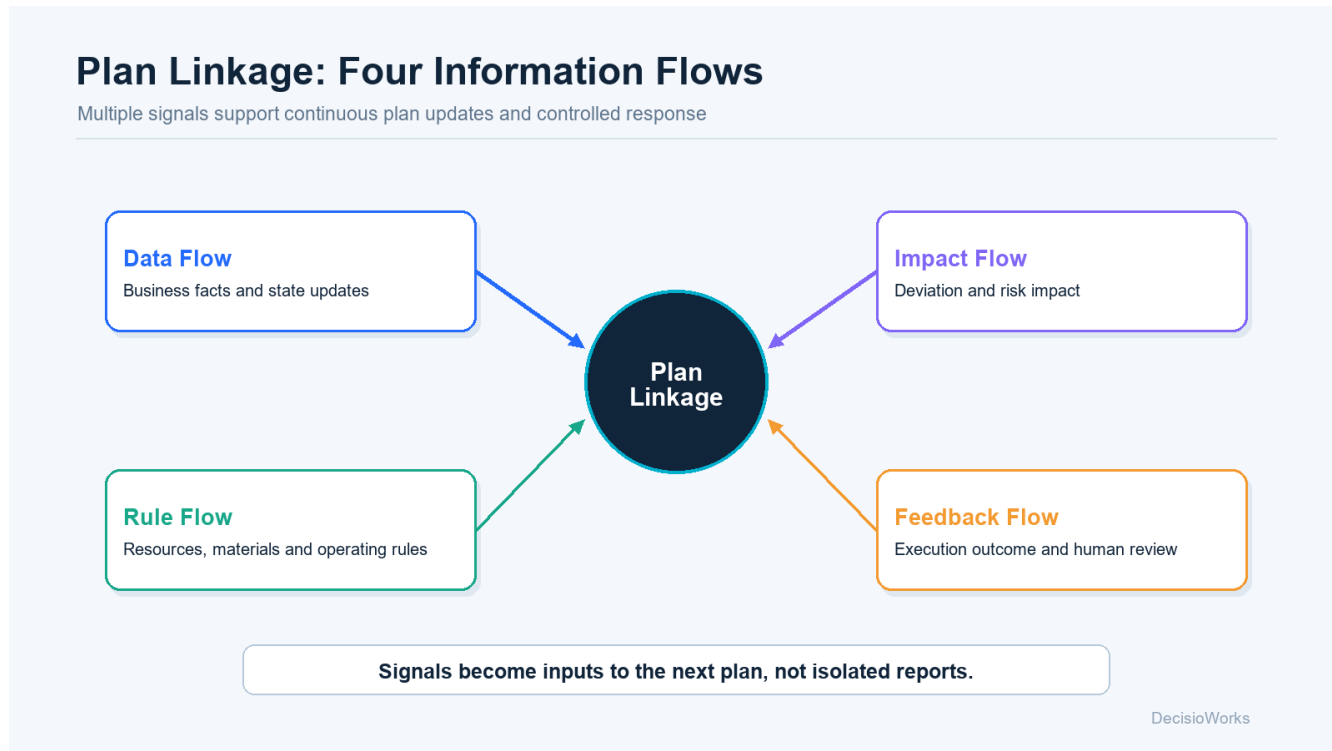


Figure 9 Plan Linkage: Four Information Flows

Underproduction, overloading, material-readiness risks, waiting, switching and execution deviations discovered by the analysis can be formed into PlanBias, which can then be converted into PlanSignal. Signals can be used for target adjustments, local rules, planned releases, or next round of data preparation. Subsequent actions will be performed according to established authority and review procedures, and the source of the signal will be retained.

Continuously Evolving Recording Requirements

Data version, mapping rules, units and responsible persons;
Target direction, weight, target and reasons for adjustment;
Source, scope, validity period and approver of site rules;
Recipes, capability versions, solver parameters and running status;
Reasons for manually accepting, rejecting or modifying the plan;
Conditions that require re-verification after changes in the business environment.

Evolutionary way

The stable part is retained as data templates, standard actions, scenario recipes and model adaptation; the changing part enters the next round through versions, rules and feedback.

SECTION 19

19 Typical application scenarios and stage value

production decision capabilities can start with the company's clearest current problems. The first round of work does not have to cover the entire plant, but it should produce results that can be inspected and support the specific judgment of "continue, adjust or stop".

Current status	first round of work	value formed	Next step judgment
Data is scattered and the relationship is unclear	Organize objects, mappings, units, and relationships	Form decision data assets and expose gaps	Can data enter the model?
Basic data is available, ROI is unclear	Select a small scene, freeze the baseline, and run recalculation	Obtain evidence of input and results	expand, adjust or stop
Already have ERP/MES/WMS	Map existing data and validate plans or schedules	See clearly the gaps in business systems and decision capabilities	Enhance existing systems
APS is online but still frequently corrected manually	Review data, goals, rules, constraints, and feedback	Locate decision chain breakpoints	Repair or rebuild local capabilities
Consulting or software partners have customer scenarios	Anonymous samples and joint verification	Convert plans into operational evidence and templates	form a joint plan
Existing models need to continuously adapt to changes	Preserve versions, configurations, deviations and reviews	Retain reusable rules and feedback mechanism	Extend scenarios and operations

The above scenarios cover data construction, direction verification, system enhancement, partner delivery and continuous operations, and are not limited to all applicable scenarios. Other scenarios can be further evaluated based on the problem scope, data preparation conditions, and result review requirements of the responsible person.

Priority

The first round of scenarios should have clear business problems, available data, responsible persons who can participate, and operational results that can be traced back to specific business objects.

SECTION 20

20 Scenario 1: High-quality production decision data assets

Suitable for enterprises whose data already exists but is scattered in ERP, MES, WMS, Excel and field records. In the first stage, data results can be formed that can be handed over, reviewed, and used for subsequent model and system construction.

First round of work scope

1. Identify orders, items, routings, operations, work centers, capacities, calendars, materials, inventory, and material-readiness objects.
2. Establish a mapping of source system fields to standard objects, and unify primary keys, enumerations, units and time coordinates.
3. Check whether the time chain, product chain, process capacity chain and material chain are closed.
4. Develop data readiness status, issue list, owners, and remediation recommendations.
5. Choose a minimal scenario to run to verify that data can enter the model and generate traceable results.

Deliverables	use
Data object and field dictionary	Unify business language and reduce repeated explanations across systems
Source system mapping table	Clarify field sources, conversions, responsibilities and update methods
Relationship and Quality Report	Expose missing, conflicting, single-unit errors and orphaned objects
Desensitization examples and test methods	Support reproduction, training and follow-up access
Baseline operational evidence	Prove that the data can not only be read, but also support the decision chain

value judgment

Even if the enterprise does not enter the complete schedule construction for the time being, clear data assets can still serve subsequent integration, analysis, model verification and system selection.

SECTION 21

21 Scenario 2: If the system direction and ROI are unclear, verify first

It is suitable for situations where the planning and scheduling issues are clear, but the enterprise is not yet sure whether to purchase APS, have self-research capabilities, enhance existing systems, or postpone investment. DecisioWorks Use real data to transform abstract solutions into comparable operational evidence.

Minimum verification scope

elements	Recommended range
scene	A production line, a bottleneck, a type of order or a planning cycle
baseline	Current plan, manual adjustments, actual results, and known issues
variable	Adjust only a few goals, rules, or resource conditions each round
index	Delivery, Load, Inventory, Switching, Waiting, Stability, and Manual Modifications
Responsible person	Planning, production, IT/data and necessary management personnel

The first round of verification focuses on answering: whether the existing data can support it, what is the scope of the problem, which constraints and goals really affect the results, whether users can review it, and what conditions are required for continued investment.

ROI Judgment needs to look at investment and reuse at the same time

Current costs of manual planning, waiting, rework, and frequent adjustments;
Investment in data preparation, authorization, integration, training and ongoing maintenance;
Indicator changes and who they affect, rather than just looking at a total score;
Whether the verified data, rules, templates and interfaces can be used in subsequent scenarios;
Whether the organization is willing to adjust the existing decision process and arrange for personnel to continue to use and review it.

Validate value

When key data are unavailable, scenarios cannot be stably defined, or evaluation criteria are unclear, the expansion of investment can be suspended, existing results retained, and re-evaluated after conditions improve.

SECTION 22

22 Scenario 3: Enhance the decision capabilities of ERP, MES, and WMS

ERP records transactions and demand, MES records execution and equipment, and WMS records inventory and logistics; production decisions also require organizing these facts into demand, capacity, process, materials, time, goals, and rules.

Typical enhancement methods

Existing system	Common Basics	DecisioWorks reinforcement
ERP	Order, BOM, procurement, inventory, delivery date	Planning objects, demand priority, process and capacity semantics, plan feedback
MES	Work orders, work reports, equipment, work in progress, exceptions	Executable scheduling, resource constraints, dynamic recalculation and deviation reflow
WMS	Inventory, batch, location, inbound and outbound	Homogeneous relationships, process-level material requirements and planning impacts
Industry software	Industry interface, process, customer accumulation	Modeled decision, controlled solving, analytical evidence, and capability orchestration

Partners can retain their own product entry points, account permissions, industry processes and customer delivery, and DecisioWorks provides standard data, model solving, result analysis and feedback capabilities. This avoids building a complete APS from scratch and does not require customers to replace existing business systems.

Key points for integration acceptance

- Whether the mapping of source systems to standard objects is stable and versionable;
- Whether the call passes the formal API, standard action and authorization path;
- Whether the results can be returned to the partner interface and recognized by business objects;
- Whether exception, timeout, no solution and authorization status are clearly distinguished;
- Whether customer-specific rules are maintained at the partner or scenario level and decoupled from the common core.

SECTION 23

23 Scenario 4: Strengthening consulting and system integration delivery capabilities

Consulting agencies and system integrators can further carry out data diagnosis, model verification and effect evaluation based on customer business understanding. DecisioWorks provides operation and analysis capabilities to support the transformation of solution recommendations into comparable and reviewable verification results.

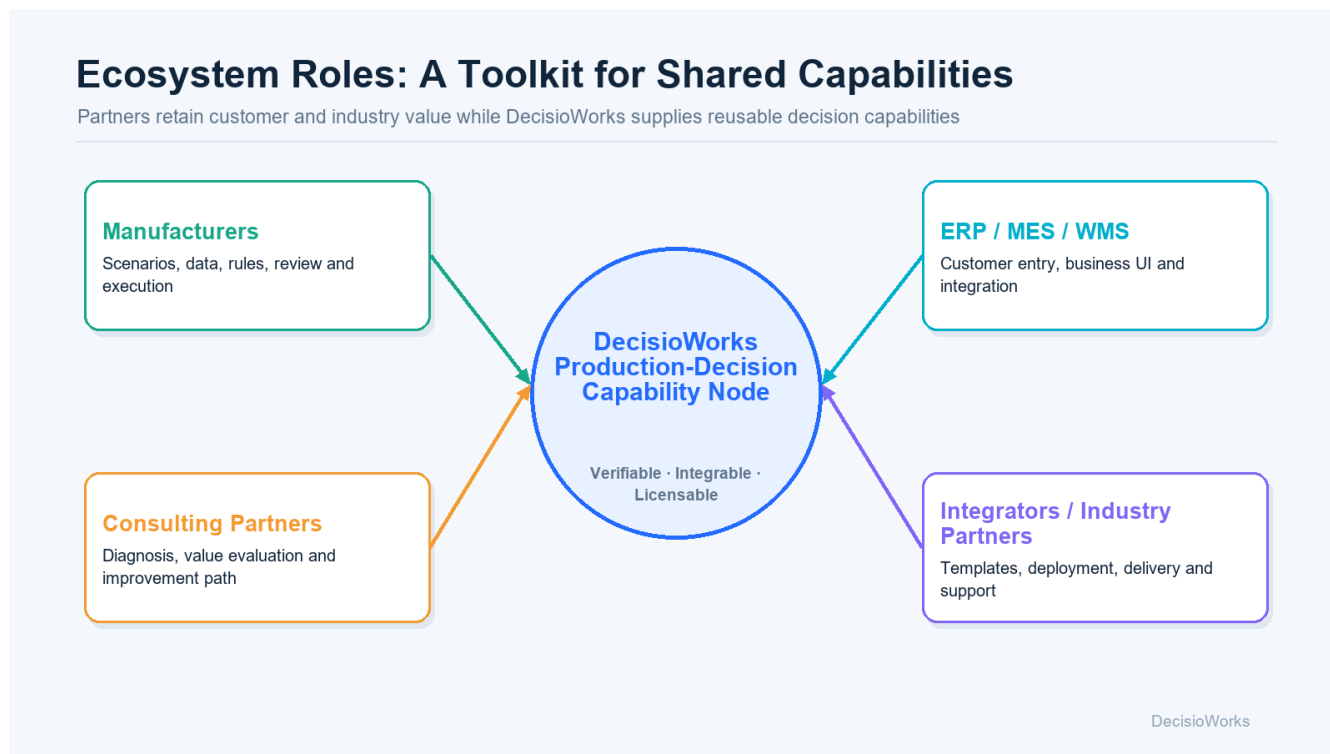


Figure 10 Division of responsibilities between DecisioWorks and ecosystem partners

Partner work	Toolkit support	The result of the union
Diagnosis of current situation	Data templates, relationship checks, readiness status	Data Gap and Capability Gap Reporting
Scheme design	Catalog of goals, rules, actions and model capabilities	Runnable scenario recipes
Effect evaluation	Baseline, adjustment recalculation, indicator and object drill-down	Before and after comparison and business review materials
ROI judgment	Inputs, Reuse, Risks and Adoption Quality	Stage recommendations for continuing, adjusting or stopping
System delivery	API, adapters, authorizations, operational status and documentation	Embedded production decision capabilities

The capabilities of both parties together form a complete solution for customers: the partner is responsible for customer relations, industry understanding, system integration and on-site services, while DecisioWorks provides common, stable, reusable production decision capabilities and agreed product support.

Partner value

The toolkit reduces the duplication of development of common capabilities, allowing consulting and delivery personnel to focus on scenario analysis, business trade-offs, and application effectiveness evaluation.

SECTION 24

24 Scenario 5: Self-construction and expansion of manufacturing enterprise IT team

Manufacturing companies with IT, data or algorithm capabilities can build and continuously expand production decision applications based on the toolkit. Enterprises retain business process, data and configuration management capabilities, while reusing model solving, standard interfaces and result analysis capabilities, reducing the workload of development from scratch.

Possible layering methods

Enterprise autonomous management	DecisioWorks provides the basis	Joint acceptance
Business objects, data responsibilities, source system interfaces	Standard data model, mapping reference and verification mechanism	Data readiness and version traceability
Goals, rules, scenario flow and interface	Target system, rule control, standard actions and recipes	The configuration source is consistent with the running behavior
Industry model selection and business evaluation	GOCK, model adaptation and controlled solving	Model state, resource boundaries, and result semantics
Production usage, permissions and change management	Web/API, logs, diagnostics and support documentation	Manual review, exception handling and revalidation

DecisioCore source code is openly available and can be used as an official reference implementation for business orchestration and integration; GOCK provides core model-based solving capabilities through authorization and adaptation interfaces. Enterprise IT teams can build and continuously scale production decision applications.

Suitable for self-build

Enterprises can designate long-term data and business owners;
 Willing to maintain version, configuration, testing and operational records;
 Key rules and organizational trade-offs can be stated explicitly;
 Start with a local scenario rather than covering the entire shop floor at once;
 Based on the common capabilities of the reused toolkit, establish the enterprise's own application operation capabilities.

SECTION 25

25 Scenario 6: Partial APS remediation and enhancement

Just because APS is online but plans to frequently return to manual work does not necessarily mean that the entire system should be replaced. A more prudent approach is to locate breakpoints in the decision chain around a problem scope.

Find clues from manual corrections

Which orders are often dragged, whether due to priority, delivery date or material not entering the model;
 Which resources are continuously overloaded, and whether they are covered up by average load or wrong capacity definition;
 Which scheduling results are always rearranged by the planner, and whether the implicit rules have not been expressed for a long time;
 Which goals conflict with on-site trade-offs, and whether the system only displays total points and not costs;
 Which execution deviations are not carried back to the next round, causing the system to keep repeating the same mismatches.

Diagnostic conclusion	priority action
Data semantics or relational issues	Fix mapping, units, time coordinates and object relationships
Goals and Rules Questions	Separate structural constraints, management goals and site rules
Model responsibilities are too large	Reduce the scenario and split the planning layer and scheduling layer
The results are uninterpretable	Added object drill-down, target material and baseline comparison
Missing feedback	Establish deviations, signals, versions and revalidation mechanisms

Reinforcement Principle

Repair local links that can be clearly positioned first; only when the data, models, and organizational processes are not suitable as a whole, consider rebuilding on a larger scale.

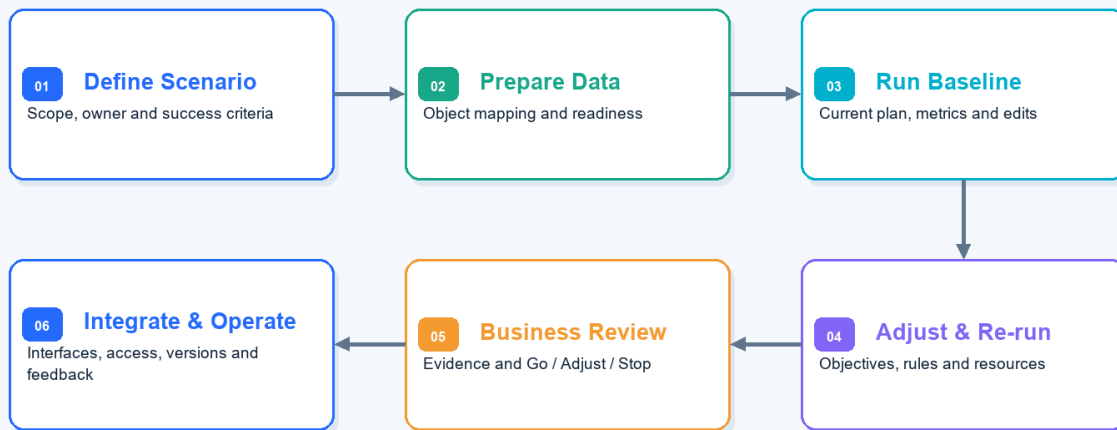
SECTION 26

26 End-to-end implementation method

Enterprises or ecosystem partners can adopt the toolkit according to the following steps. Each stage will form a checkable output and decide to continue, adjust or stop based on this. The implementation process starts with problem identification and gradually advances to operational verification, system integration and continuous operation.

From Scenario to Operations

A six-stage implementation path with reviewable outputs at every stage



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Figure 11 Six-stage implementation route from scenario definition to integrated operation

stage	core activities	main output
scenario definition	Clarify the problem, scope, cycle, responsibilities and success criteria	Scenario Description and Boundary List
Data preparation	Object mapping, relationship verification, unit and version confirmation	Mapping table, readiness status and issue list
Baseline run	Reproduce the current plan and manual adjustments	Baseline configuration, results and business metrics
Adjustment recalculation	Change a small number of goals, rules, or resource parameters	Program differences and affected objects
Business review	Interpret results, confirm enforceability and trade-offs	Phase conclusion of continuing, adjusting or stopping
Integrated operations	Access the system, set permissions, logs, support and feedback	Formal interface, runbook and iteration checklist

Implementation cycle and workload depend on data quality, scenario complexity, source system interfaces, model adaptation, and organizational involvement. Enterprises and implementation partners can develop milestones, acceptance criteria, and change mechanisms after the scenario is confirmed to gradually promote application.

SECTION 27

27 Pilot selection and data preparation

The first round of pilots should choose scenarios with clear scope, available data, and comparable indicators, so that the business and technical teams can form clear verification conclusions within the agreed period.

Five conditions for qualified pilot

1. Business issues can be clearly stated: such as delivery risks, bottleneck loads, excessive model changes, or package mismatches.
2. Data can be obtained: at least covering requirements, products, processes, resources, time and applicable material conditions.
3. Responsible people are able to participate: planning, production and data personnel are willing to confirm the definition and double-check the results.
4. Results can be compared: there is a current plan, historical performance or a clear threshold as a baseline.
5. Scope can be frozen: Don't change too many systems and management rules simultaneously during the first cycle.

Data category	minimal content	Common supplements
need	Order, product, quantity, delivery time, priority	Customer level, frozen status, split rules
Craftsmanship	Route, process, sequence, cycle time or productivity	Alternative routes, lead time, batch size
resource	Work center, capacity, shift, calendar	Maintenance, sharing of resources, personnel and molds
materials	BOM, inventory or packaged information	Material arrival time, substitute materials, process level consumption
Baseline and Target	Current plan, manual adjustments, evaluation indicators	Historical execution deviations and reasons

data principles

The pilot phase does not require the opening of the complete database, but only prepares minimal, redacted, interpretable, and replayable data packages around the scenarios and fields confirmed by both parties.

SECTION 28

28 Deliverables and Acceptance Criteria

The following checklist helps enterprises or implementation partners check the scenario results, covering data, configuration, operation records and business review, to facilitate phase acceptance, problem tracking and subsequent expansion.

Deliverables	minimum content	Acceptance issues
Scenario description	Problem, scope, cycle, responsible person, success criteria	Whether it is possible to clearly judge whether it is applicable or not
Data mapping and readiness reporting	Object, field, unit, relationship, version, gap	Whether the agreed ready state is reached
Goals and rules list	Direction, weight, scope, source, validity period	Is it configurable, traceable, and free of hidden conflicts?
Run recipes and configurations	Actions, dependencies, parameters, resources and authorization requirements	Can it be reproduced according to the official path?
Results and evidence	Overall indicators, object details, audit materials, diagnostic information	Can you explain the solutions and differences?
Review and restrictions	Reasons for acceptance, rejection, modification, and known limitations	Whether to support the next round of adjustments
Integration and operation and maintenance materials	API, deployment, permissions, logs, support and upgrades	Whether it can be operated and handed over for a long time

Multidimensional acceptance requirements

Runnable: Able to execute stably under the target environment and configuration;

Interpretable: able to explain bottlenecks, conflicts, constraint hits, and trade-offs;

reviewable: The responsible person can drill down to specific objects and form conclusions;

Repeatable: data, configuration, version and running status are traceable;

Extensible: Content that passes verification can move into integration and the next scenario.

evidence boundaries

Verification can start with public sample drills and gradually advance to enterprise real data verification and production application acceptance, with corresponding results and review records saved at each stage.

SECTION 29

29 Integration patterns and deployment topologies

DecisioWorks can run stand-alone or embedded into existing industrial software or partner solutions. Integration design should first confirm the data flow, call flow, result flow and responsibility boundaries, and then select the file, database, API or application embedded method.

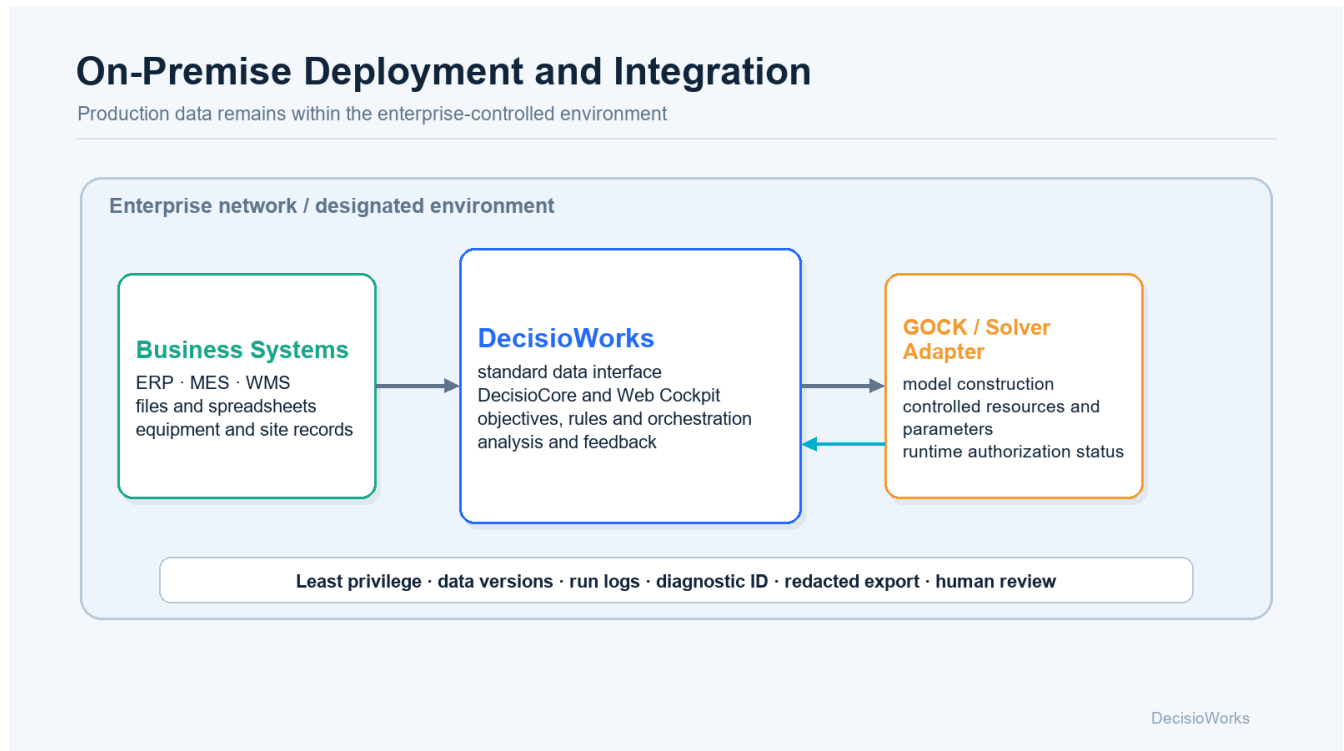


Figure 12 Local deployment and system integration topology

Integrated mode	Suitable for stage	Features
File/packet verification	First-round scenarios and anonymous examples	Quick preparation, clear scope, easy playback
Database or batch interface	Periodic planning and offline operation	Suitable for stable batch and versioned input and output
API call	Business system embedded and on-demand operation	Requires permissions, error codes, idempotence and state management
Partner app embed	Industry-specific products or delivery solutions	Partners retain interfaces and processes, and call toolkit capabilities
AI Agent call	Configuration assistance, process triggering and result interpretation	Only registered capabilities are called, and high-impact actions are manually reviewed.

DecisioCore assumes responsibilities before and after solving: organizing data, goals and rules before solving, and restoring plan results into business indicators, object details and feedback signals after solving. GOCK receives explicit model calls through the adapter and does not directly expose the internal implementation to the upper-layer business.

SECTION 30

30 Data security and operation management

Production data, planning results and authorization information may affect business operations. The solution emphasizes local and controlled operation, least privileges, versioned configuration and traceable logs by default.

control domain	Basic requirements	suggest evidence
data	Clarify storage location, access roles, backup, retention and redaction rules	Data list, permission records, redaction examples
network	Confirm access boundaries between web, API, database, and external systems	Deployment topology, firewall, or interface instructions
Authorize	Authorization documents are kept under controlled conditions, with clear scope of capabilities, time and resources.	Authorization status, update records, error codes
run	Distinguish between success, timeout, no feasible solution, component and authorization exceptions	Run logs, diagnostic IDs, versions and parameters
support	Issue requests contain only necessary, redacted, reproducible material	Support tickets, reproduction steps, processing records
decision making	High-impact results are reviewed by designated responsible persons	Acceptance, rejection, modification and approval records

DecisioWorks commercial version supports local deployment and controls corresponding capabilities through authorization files; the standard local operation process does not require uploading business data to the external cloud. When enabling external AI, remote support, or third-party interfaces, access permissions, data flow, and redaction methods can be configured according to enterprise data management requirements.

Plan review and business application

DecisioWorks supports data preparation, scenario generation, comparison, interpretation and review. Planning, production and management personnel can combine site status, quality requirements and customer commitments to select appropriate solutions and incorporate them into the company's existing approval and execution processes.

Version management and continuous verification

Records of version updates, configuration adjustments, and model extensions should be kept to facilitate comparison of before and after results, locate the source of changes, and reuse verified configurations in subsequent runs.

Baseline verification should be re-performed after important changes to the data structure, business rules, model version, operating environment, or applicable scope.

SECTION 31

31 Continuous adoption path between research version and commercial version

The research version and the commercial version share data semantics, action organization, and usage logic. The research version helps users learn, evaluate and conduct non-production verification; the commercial version supports actual business applications, partner integration and continuous operation through corresponding authorizations.

01 Research and understanding	02 Version selection	03 Scenario verification	04 System integration	05 Continued operations
Read the documentation, samples, and research version descriptions to clarify the business problem and data scope	Select the version according to application requirements, configure capabilities, resources and product support	Freeze the baseline and adjust recalculation. After business review, decide to continue, adjust or stop.	Confirm the new usage scope. Connect to the enterprise or partner system and configure permissions and feedback.	Maintain data, rules and templates and re-verify after business or version changes

Figure 13 Research understanding, version selection, scenario verification and continuous operation path

stage	fit for purpose	Things to note
Research Edition	Learning assessment, scenario preparation and non-production verification	Suitable for understanding capabilities, becoming familiar with data requirements and practicing operational procedures
Minimal verification	Baseline and recalculate evidence using real or masked data	Clarify the scenario scope, evaluation indicators, participants and operating conditions
Commercial Edition	On-premises deployment, formal capability authorization and controlled operation	Select capabilities, operational resources and product support based on business needs
System integration	Embed in enterprise systems or partner applications	Requires interfaces, permissions, exceptions and operation and maintenance design
continuing operations	Long-term maintenance of versions, templates, rules and feedback	Revalidate after major changes

The product homepage summarizes versions, documents, demonstrations and cooperation entrance: <https://qiaoyx-or.github.io/production-decision-sandbox/>. The official/commercial version release address is <https://gitee.com/qiaoyx-or/decisioworks>, and the research version release address is <https://gitee.com/qiaoyx-or/decisioworks-research>.

Adopt principles

Understand the capabilities from the research version, and then select the version and authorization scheme according to application requirements; accumulate operation results through scenario verification, and gradually enter system integration and continuous operation.

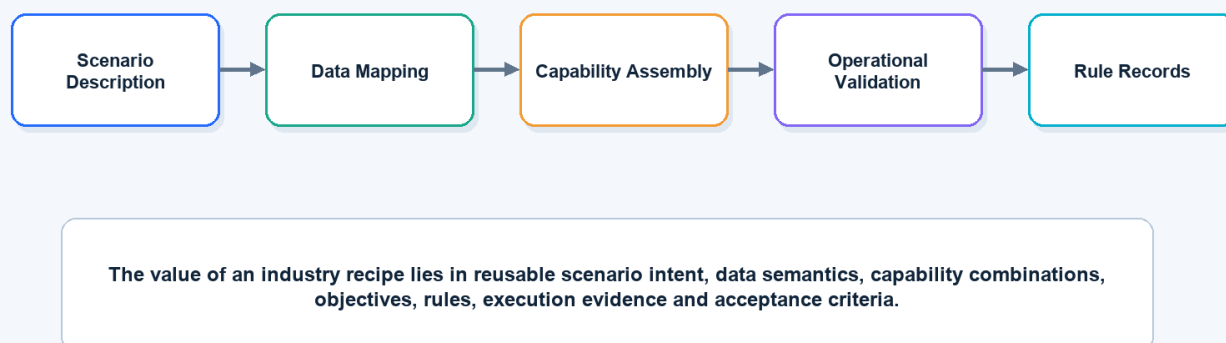
SECTION 32

32 Ecosystem division of labor and partner access

DecisioWorks Complements capabilities with industry partners: the toolkit provides common model-based decision capabilities, and partners combine industry knowledge, business processes and customer services to build application solutions suitable for specific industry scenarios.

Turning Industry Recipes into Reusable Implementation Assets

A recipe combines scenario logic, data mapping, capability registration, execution and review



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Figure 14 Industry recipes move from sample collections to reusable implementation assets

Role	Main contributions	Ability suitable for use
manufacturing company	Scenarios, data, rules, goals and execution review	Scenario experiments, target rules, and result analysis
consulting agency	Problem diagnosis, solution comparison and improvement paths	Data templates, baseline recalculation, evidence reporting
system integrator	ERP/MES/WMS access, deployment and operation and maintenance	Data mapping, API, standard actions and authorization
Industry software partner	Industry interfaces, processes and customer applications	Capability registration, recipe, model adaptation
Park and industrial platform	Demand collection, enterprise services and pilot coordination	Research Edition, scenario checklist and validation framework
AI Agent Developer	Configuration draft, process invocation and explanation	Capability Catalog, Standard Actions and Operational Evidence

Four stages of partner access

1. Scenario validation: Define business requirements into verifiable decision problems.
2. Data and Capability Preparation: Complete source data mapping, readiness checks, and capability selection.
3. Baseline and adjustment verification: form before and after comparisons and business review records.
4. Integration and operations: establishing interfaces, versions, permissions, monitoring, support and continuous iteration.

SECTION 33

33 Value Indicators and Verification Framework

The value of a production decision cannot be summarized by just a composite score or an efficiency ratio. At least four dimensions should be observed simultaneously: data assets, business results, operational quality, and organizational adoption.

Value Validation: Look Beyond a Single Score

Compare baseline and adjusted scenarios with the same dataset and measurement definitions

Dimension	Questions and metrics	Required evidence	Decision use
Data asset	Objects, relations, units, completeness	Mapping, validation report, version	Ready for modeling?
Business outcome	Delivery, load, inventory, changeovers, waiting	Metric details and object drill-down	Worth adopting?
Run quality	Feasibility, status, duration, operating limits	Run log, exceptions and diagnostics	Capability stable?
Adoption quality	Human changes, reasons and owner	Review record and adoption decision	Sustainable use?

The conclusion is a traceable Go, Adjust or Stop decision, not simply a higher score.

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Figure 15 Multidimensional value verification matrix

Dimensions	Example indicators	Evaluation definition
data assets	Mapping overrides, relationship errors, unit conflicts, ready status	Object scope, data version and responsible person
business results	meet, shortage, delay, load, inventory, switch, wait	Units, granularity, baselines, and affected objects
running quality	Feasibility, status, time consumption, parameters and resource boundaries	The difference between timeout, no feasible solution and exception
Adoption quality	Manually modified items, modification reasons, acceptance rate, review completion degree	Responsibility, rationale and implementation feedback

Verification should use the same data and the same set of evaluation criteria to compare the baseline and the adjustment plan. Only a few factors are changed each round, and what is improved is recorded, at what cost, who is affected, and whether new risks are introduced.

Indicator principles

Any business indicators should be indicated with units and granularity, and be able to drill down to specific orders, resources, operations or time units.

SECTION 34

34 Investment return evaluation and stage-gate decisions

The investment return assessment (ROI) should be carried out based on the construction investment, operating effects, asset reuse and organizational adoption in real scenarios. As the verification evidence gradually improves, companies can update their revenue expectations and investment plans to formulate staged decisions.

ROI composition

category	What needs to be recorded
current cost	Manual planning, overtime coordination, waiting, rework, frequent adjustments and delivery risks
Construction investment	Data preparation, software licensing, integration, training, maintenance and organizational changes
Changes in results	Delivery, Load, Inventory, Switching, Waiting, Stability and Impact Objects
Reuse value	Data templates, field mappings, rules, recipes, interfaces and training materials
Adoption Risks	Data responsibility, user acceptance, process access, model failure conditions and support capabilities

Three types of stage conclusions

in conclusion	Judgment conditions	Follow-up actions
keep pushing forward	Data relationships are stable, results can be drilled down, and the recognition of responsible persons continues to expand.	Enter system integration or expansion scenarios to improve data, processes and operational support
Adjust validation	The run has completed, but data, goals, rules, or scope still need to be added	Keep a list of issues and move on to the next round of small-step adjustments
Suspend investment	Boundaries are not stable, key data are not available, or evaluation responsibilities are missing	Stop expanding investment and preserve existing assets and conclusions

decision value

The significance of low-cost verification is not to ensure that the project will continue, but to allow continuation, adjustment and stopping to be based on traceable evidence.

SECTION 35

35 Risk List and Governance Mechanism

production decision solutions involve business, data, models, software and organizational collaboration. Risk management runs through scenario confirmation, data preparation, operational verification, system integration and continuous operations, and is implemented through division of responsibilities, inspection records and change mechanisms.

risk	Performance	Governance style
The scene is too big	The first pilot covers too many workshops and targets	Narrow it down to a freezable and reviewable range
Insufficient data readiness	The fields are complete but the object relationships and units are wrong	Five-level readiness, four relationship chains and real operation
Rules are not explicitly expressed	Artificial experiences are scattered among scripts, meetings, and personal memories	Rule source, scope, validity period and approval record
Target misreading	Treat the weight or total score as actual business volume	Target material auditing and object-level metrics
The results were not fully reviewed	Only look at the optimal solution, not the cost and applicable conditions	Candidate comparison, interpretation review and manual validation
integrated coupling	Customer-specific rules are coupled with the common core, making upgrades more difficult	Data, business, model, adaptation and interface layering
Authorization and security	Capability override, parameter overrun, data leakage	Local deployment, least privilege, formal authorization and redaction support
organizational mismatch	No responsible person or planning process accepts the results	Confirm responsibilities before piloting and record reasons for acceptance and modification

Constraint Relaxation Is Governed Problem Solving

An infeasible result can reveal conflicts among business facts, objectives and operating rules



Relaxation does not mean deleting a rule. The conflict, adjustment candidate, business cost, approval and effective period must remain visible.

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Figure 16 Auditing and governance of infeasible issues

Relaxing a constraint does not mean removing the constraint. It should locate conflicts, formulate candidate adjustments, assess costs, and be confirmed by those responsible. The absence of a feasible solution is also an important business signal in itself, which may indicate that the current constraints, resources, and goals cannot be established at the same time.

SECTION 36

36 Example: Injection Scheduling and Type Change Decisions

This section uses a public injection molding example to illustrate the method, focusing on showing how color, cap type, capacity, batch size and resource cycle enter into the changeover decision, and understanding the target trade-offs through baselines and recalculations.

elements	Sample expression	Acceptance issues
Business attributes	Color, lid type, and container capacity are mapped to clear semantics	Whether to avoid relying on ambiguous attribute slots
Resources and cycles	2 work centers, sequence length and number of cycles	Are the task scope and cycle connections reasonable?
constraint	Capacity, demand, inventory, sequence switching and cycle switching	Whether the constraint is enabled according to the business scope
Target	Relative trade-offs between color switching and capacity status changes	Whether the weight enters the target material as configured
result	Sequence, requirement coverage, switch, batch and audit material	Can changes and costs be explained?

Verification method

1. Fix the same data version and scenario recipe and run the baseline.
2. Only a few weights, sequence lengths, or loop conditions are adjusted per round.
3. Compare sequence switches, cycle switches, requirement coverage, and affected objects.
4. Check the target material direction and do not interpret the weighted value as actual business volume.
5. It is up to the person in charge of the plan or process to judge whether the plan is executable.

The significance of public samples is to prove that the link is operational, the configuration is auditable, and the results are comparable. The customer environment still needs to re-confirm equipment, processes, batches, model change logic, solution resources and business acceptance criteria.

Example conclusion

Increasing the weight of a certain goal does not mean that the corresponding business indicator will necessarily improve in one direction; the result is determined by the data, constraints, other goals, and the solution structure.

SECTION 37

37 Example: stamping multi-process planning chain

This section uses a public stamping example to show the complete path from order, process, capacity and material-readiness data to controlled planning, result interpretation and target audit.

link	Sample highlights	Need to check
enter	Order details, process routes, process-level materials, kits, capacity and time units	Object relationships, time coordinates, units and existing occupancy
Model	Capacity, demand, material-readiness constraints and delivery/waiting targets	Hard and soft constraints, target direction and applicable conditions
process	Data collection, constraint resolution, generation, selection, evaluation and optional dispatch	The input, output and status of each action
result	Output, Shortage, Load, Waiting and Target Materials	Whether the overall indicator can drill down to the object
Recalculate	Keep data consistent with resources and only adjust a few target parameters	Whether the plan changes and whether the target materials change according to the configuration

Key points for business review of planned results

Requirements coverage: whether there are unscheduled, duplicated or inconsistent time coordinates;

Work center load: whether local bottlenecks are masked by the overall average;

Materials are complete: in which process and time unit the risk occurs;

Plan stability: which objects remain unchanged and which move after adjusting the target;

Target material: whether weight changes are included in the calculation, and whether the actual business volume changes simultaneously;

Manual review: Whether the results comply with on-site execution, delivery and risk commitments.

The example may have a situation where "the target weight changes, but the plan results remain stable". This may indicate that the same solution is still dominant under the current data and constraints. During review, target materials and actual business indicators should be checked at the same time, and configuration changes and program changes should be distinguished.

SECTION 38

38 Phased construction roadmap

The construction of production decision capabilities is suitable for gradual expansion from local verification. Each stage forms assets and provides entry conditions for the next stage, rather than treating the preliminary work as a one-time consumption.

stage	core goals	Main assets	Conditions for entering the next stage
Phase 1: Observation and Preparation	Understand the boundaries of capabilities and choose scenarios	Scenario list, data sample, responsible person	Problems can be defined and data available
Phase 2: Data Asset Formation	Establish objects, semantics, relationships and quality status	Map, Dictionary, Checksum Version	Meet the agreement ModelReady/SolverReady
Phase 3: Run Verification	Baseline, recalculate and business review	Configuration, results, evidence and conclusions	DecisionReady, recognized by the responsible person
Phase 4: Business Integration	Access to business systems and formal support	API, deployment, authorization, operation and maintenance and training	Stable operation and clear service boundaries
Phase 5: Ongoing operations	Retain rules, templates, feedback and expand scenarios	Recipe library, version, review and re-validation mechanism	Organizational development of long-term capabilities

From Run to Feedback: The Production-Decision Loop

Data, objectives, solving, review and feedback remain connected across cycles



DecisioWorks

Figure 17 Closed loop of production decision from operation to feedback

After the company completes the verification of local scenarios, it can gradually expand to other workshops, planning levels or partner systems. Save data, configurations, results and review records for each round, allowing proven experience to become the starting point for the next phase.

SECTION 39

39 Starting point and next step of cooperation

The first cooperation can start with a specific production plan, scheduling or resource allocation problem, clarify the data scope, verification goals and review personnel, and establish a common basis for subsequent operations and integration.

Recommended first communication materials

Brief description of the scenario: How is it currently planned and what are the most common manual adjustments;
 Scope: workshop, production line, product, planning cycle and time granularity;
 Data list: orders, processes, resources, calendar, inventory/package and current schedule;
 Goals and rules: the most important indicators, non-violable conditions and on-site experience;
 Use of results: who will review it, how to compare it, and whether it is necessary to write back the existing system;
 Security requirements: redaction, deployment, access, logging and retention periods.

Simplify the cooperation process

Initial communication → Scenario confirmation → Data preparation → Operational validation → Result review → Integration extension assessment. In the first round, the participants, verification goals, time schedule and subsequent promotion conditions will be jointly confirmed.

public entrance	address
Product Home	https://qiaoyx-or.github.io/production-decision-sandbox
Official version/commercial version	https://gitee.com/qiaoyx-or/decisioworks
Research Edition	https://gitee.com/qiaoyx-or/decisioworks-research
Contact email	qiaoyuxuan@decisioworks.com
Official account	AI + production decisions

principle of cooperation

Conduct verification with clear scenarios, support assessment with operational evidence, and then determine subsequent integration scope and cooperation methods based on business review results.

SECTION A

A Appendix A: List of standard data objects

The following objects are used to help enterprises prepare the first round of data and do not require all content to be provided at once for all scenarios. Specific scope should be determined by program level, business issues, and applicable constraints.

object group	core object	key relationship
Requirements and products	Order, order details, products, priority, delivery date	Order points to product and demand time
Craftsmanship	Process route, process, sequence, batch size, rhythm/productivity	Products enter resources through routes and processes
resource	Work center, equipment group, capacity, shift, calendar, maintenance	Resource capabilities are mapped to time units
materials	Materials, BOM, process level usage, inventory, material readiness	Material conditions bind products, processes and times
Goals and rules	Target items, direction, weight, site rules, validity period	Acts on objects, scopes and decision contexts
plan results	Planned volume, start/end, resources, status, deviation	Traceable to requirements, operations and time units
operational evidence	Data version, recipe, parameters, status, diagnostic ID	Connect the input, process, and results of a run

minimal relation check

- Whether valid products can be found for each requirement;
- Whether a suitable process route can be found for each product;
- Whether executable resources can be found for each process;
- Whether the resource has capacity and calendar covering the planning period;
- In applicable scenarios, whether the material requirements can be found in inventory or material readiness of information;
- Whether the planning results can be returned to requirements, resources, processes and time.

SECTION B

B Appendix B: First Round Pilot Checklist

The following checklist can be used for the first scenario validation meeting and data preparation. Tick results should be saved along with the list of issues, responsible persons and planned dates.

Check items	Confirmed	Responsible person/description
Pilot issues, scope, cycle and success criteria are defined	☐	
Planning, production, IT/data owners identified	☐	
Demand, process, resource and time data can be provided	☐	
Applicable material, inventory or material-readiness data can be provided	☐	
Data sources, units, enumerations and update methods have been explained	☐	
The current plan and manual adjustments can be used as a baseline	☐	
Structural constraints, management objectives and site rules are distinguished	☐	
The units, granularity and thresholds of the evaluation indicators have been determined	☐	
Result review personnel and review time have been arranged	☐	
Desensitization, deployment, access and data retention requirements confirmed	☐	
Formal authorization, operational resources and support boundaries confirmed	☐	
Conditions for expansion, adjustment and cessation have been agreed upon	☐	

Usage suggestions

Checklists are used to clarify pilot readiness status. Unconfirmed items should be recorded with the responsible person, supplementary tasks and completion time; items involving business assumptions or default values should be confirmed before operation.

SECTION C

C Appendix C: Glossary

The following terms correspond to the engineering concepts in this white paper to facilitate understanding of component responsibilities, data requirements, and operating processes.

the term	illustrate
APS	Advanced planning and scheduling, typically covering planning issues such as demand, capacity, materials, and job sequencing
production decisions	The process of forming, comparing, reviewing and providing feedback on plans based on requirements, resources, materials, time and organizational goals
DecisioWorks	An open capability toolkit for manufacturing production planning, job scheduling and collaborative decision
DecisioCore	Open reference implementation for decision orchestration and integration, connecting data, goals, rules, standard actions, model results and feedback
GOCK	General-purpose optimization calculation core, providing model construction, target constraint expression and controlled solving capabilities
Model-Data Resonance	Models and data are calibrated with each other in the same capability chain and continue to evolve through result feedback
DecisionContext	Describe the scenarios, inputs, goals, rules, responsibilities, and audit context of a decision run
marginal_control	Undertake the rule control mechanism of on-site rules, bottleneck protection, priority windows, prohibition conditions and resource quotas
Ability orchestration recipe	Scenario configuration describing standard actions, parameters, dependencies, execution methods and acceptance conditions
data specification	Define data access conventions for business objects, field semantics, units, relationships, constraints and versions
SolverReady	Data, goals, rules, and configurations form a state of readiness to solve the problem
DecisionReady	Results can be interpreted, compared, reviewed and made ready for organizational judgment

SECTION R

R References

The following information provides industry scenarios, technical standards and related solution information for further reading. Data access date: September 6, 2026.

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- [11] DecisioWorks Wiki (Chinese), <https://github.com/qiaoyx-or/DecisioWorks.wiki>
- [12] DecisioWorks product homepage, <https://qiaoyx-or.github.io/production-decision-sandbox>

Supporting documentation and product information

The product homepage and Wiki provide versions, samples and usage documents to support product selection, scenario verification and system integration.

Release Notes

The specific version functions are subject to the official release notes, and the scope of authorization, service content, and rights and responsibilities of both parties are agreed upon in the formal agreement.